

Comprehensive System Manual & Banking Policy Guide

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Executive Summary: This document establishes the authoritative operational guidelines, fee structures, compliance requirements, transactional limits, security controls, and end-user system navigation procedures for Nova Digital Bank. It serves as both the canonical reference for customer accounts and the knowledge base for Nova's AI Banking Assistant.

1. Customer Account Architecture & Account Types

Nova Bank provides three primary tiers of digital retail and commercial accounts. Every account is provisioned with an international IBAN, routing transit number, and cryptographic ledger tracking.

Account Tier	Target Customer	Minimum Deposit	Monthly Maintenance	Standard APY
Standard Checking	Everyday transactions, salary deposits	.00	.00 (Free)	0.10%
High-Yield Savings	Wealth accumulation, emergency fund	.00	.00 (Free)	4.75% Compounded Daily
Premium Wealth Tier	High net-worth & multi-currency holders	,000.00	.00 (Waived at +)	5.20% APY + VIP Perks

2. Complete Fee Schedule & Transaction Limits

All fees are evaluated and settled automatically in real time according to the following schedule:

Operation / Service	Standard Tier Fee	Processing Window	Daily Limit (Per User)
Internal Account Transfer (P2P)	.00 (Always Free)	Instant (Sub-second)	,000.00 / day
Domestic ACH Wire Transfer	.00 standard / .00 express	ACH: 1-2 days / Same-day	,000.00 / day
International SWIFT Outbound	.00 flat + 0.3% FX spread	1 to 3 Business Days	,000.00 / day
ATM Cash Withdrawal (In-Network)	.00 (All 55k+ Allpoint ATMs)	Instant Cash Dispense	,500.00 / day
Overdraft Cushion Protection	.00 up to -.00 cushion	Immediate Auto-coverage	Max -.00 balance

3. Security Architecture, Cards & Account Freezing

Card Management & Virtual Number Generation:

- Instant Virtual Cards:** Users may generate up to 5 disposable virtual cards per calendar month directly from the Cards dashboard. Virtual cards support custom single-transaction and monthly caps for secure e-commerce.
- Physical Debit Cards:** Shipped via tracked courier within 3-5 business days upon account creation. Supports contactless NFC and EMV chip PIN.
- Emergency Account & Card Freeze:** If a card is lost or unauthenticated charges occur, users or support personnel can issue an instant freeze command. A frozen card immediately rejects all authorizations while allowing inbound deposits.
- FDIC Insurance & Cryptographic Encryption:** Deposits are insured up to ,000 per depositor. Data at rest is encrypted with AES-256-GCM; communications enforce TLS 1.3.

4. Wealth, Multi-Currency Forex & Lending Formulas

Multi-Currency Engine: Nova supports real-time multi-currency wallets in USD, EUR, GBP, JPY, CAD, AUD, CHF, and SGD. Interbank mid-market exchange rates are streamed with a 0.3% fixed spread.

Personal Loan & Mortgage Underwriting Rules:

- **Borrowing Amounts:** Unsecured personal loans range from ,000 to ,000 with repayment tenors of 12, 24, 36, 48, or 60 months.
- **Interest Rate Range:** Annual Percentage Rate (APR) begins at 6.49% for prime tier credit scores (740+) and caps at 18.99% for standard credit scores.
- **Monthly Installment Formula:** Monthly payment M is calculated via standard amortization:

$$M = P * [r(1 + r)^n] / [(1 + r)^n - 1]$$
 (where P = Principal, r = Monthly Interest Rate, n = Number of Payments)

5. User Operating Guide — How to Use This System

This step-by-step operating guide details how customers and administrators navigate the Nova Banking Web Platform and interact with the AI Assistant:

Step 1: Logging In & Authentication

1. Open the Nova Banking Portal in your web browser (default URL: `http://localhost:5173`).
2. Enter your registered email (e.g. `john.doe@bank.com`) and secure password (e.g. `password123`).
3. If Multi-Factor Authentication (MFA) is enabled, enter the 6-digit TOTP code sent to your authenticator app.

Step 2: Managing Accounts & Reviewing Balances

1. Upon authentication, you will be directed to the **Dashboard Overview**, displaying your active accounts, primary checking balance, high-yield savings, and recent activity.
2. To open an additional account (e.g. High-Yield Savings or Euro Wallet), click **'Open New Account'** in the top-right header, select the currency and account type, and confirm.

Step 3: Sending Transfers & Managing Payees

1. Click **'Transfers'** on the sidebar navigation.
2. Select the **Source Account** from which funds will be deducted.
3. Enter the **Destination Account Number** or pick from your **Saved Payees** list.
4. Input the transfer amount and optional description, then click **'Review & Execute Transfer'**.

Step 4: Interacting with Nova's AI Banking Assistant

Nova integrates a fully conversational, tool-calling AI assistant powered by autonomous SubAgents and ChromaDB RAG. You can interact with the assistant via the floating chat widget on any screen:

- **Account Operations:** Type *'What is my current balance?'* or *'Show my last 5 transactions'*. The AI will autonomously call backend banking tools to fetch your real-time balances.
- **Transferring Money:** Type *'Transfer to account 100002 for dinner'*. The AI will inspect parameters, initiate the transfer, and provide an interactive confirmation receipt.
- **Security Controls:** Type *'Freeze my debit card immediately'* or *'Set my daily spending limit to '*.
- **Currency Conversion & Loan Calculations:** Type *'Convert 500 USD to EUR'* or *'Calculate monthly payment for a ,000 loan at 7.5% for 36 months'*.
- **Policy & Fee Inquiries:** Type *'What are the wire transfer fees?'* or *'How does high-yield savings APY work?'*. The AI will query the ChromaDB Vector Store and answer with citation accuracy.

Step 5: Uploading & Ingesting Documents into RAG Knowledge Base

1. Navigate to the **Knowledge Base / RAG Admin** view in the platform.
2. Click **'Upload Document'** and select a policy document or PDF (such as this manual).
3. Select your preferred chunking strategy: **Dynamic LLM Semantic Chunking** (recommended for complex multi-topic docs) or **Fixed-Window Chunking**.

4. Click '**Ingest into ChromaDB**'. The system will atomically chunk, embed, and index the document. Once completed, the AI assistant will immediately answer customer queries using the new knowledge.

Step 6: Security Best Practices & Support Escalation

- Never share your banking password, PIN, or MFA tokens with anyone.
- If you detect unauthorized access, immediately freeze your accounts via the AI chat or settings menu.
- For human support escalation, contact Nova Support 24/7 at support@novabank.com or 1-800-555-NOVA.